

Problem A. Concentration inequality

In this problem, we will derive a concentration inequality that is finer than Hoeffding's inequality.

Let $X_1, \dots, X_m$ be independent random variables drawn according to some distribution $\mathcal{D}$ with mean p and support included in $[0, 1]$. Let $q \in [0, 1]$ and define $\hat{p}$ by $\hat{p} = \frac{1}{m} \sum_{i=1}^m X_i$.

1. Use Markov's inequality to prove that for any $t > 0$,

$$\mathbb{P}[\hat{p} \geq q] \leq e^{-tmq} \prod_{i=1}^m \mathbb{E}[e^{tX_i}].$$

2. Show that

$$\mathbb{P}[\hat{p} \geq q] \leq [e^{-tq} (1 - p + e^tp)]^m$$

(Hint: you can show that $\forall x \in [0, 1], e^{tx} \leq 1 - x + e^tx$).

3. Choose $t > 0$ to minimize the right-hand side and prove the following inequality:

$$\mathbb{P}[\hat{p} \geq q] \leq e^{-mD(q||p)},$$

where $D(q||p) = q \log \frac{q}{p} + (1 - q) \log \frac{1-q}{1-p}$ is the binary relative entropy of p and q .

4. Show that this can be used to derive a bound that is finer than Hoeffding's inequality by choosing $q = p + \varepsilon$, for any $\varepsilon > 0$ (Hint: you can use Pinsker's inequality, which in this context gives $D(q||p) \geq 2|p - q|^2$).

Problem B. PAC-Learning

1. **Concentric spheres.** Let $\mathcal{X} = \mathbb{R}^3$ and consider the set of concepts of the form

$$c = \{(x, y, z) : x^2 + y^2 + z^2 \leq r^2\} \quad \text{for some } r > 0.$$

Show that this class can be (ε, δ) -PAC-learned from training data of size

$$m \geq \frac{1}{\varepsilon} \log\left(\frac{1}{\delta}\right).$$

You should carefully justify all steps of your proof. (For this problem, you should seek a proof similar to one presented in class for rectangles in the plane and not leverage a more powerful or more general theorem.)

Problem C. Rademacher complexity

1. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two families of functions mapping $\mathcal{X}$ to $[0, 1]$ and let S be a sample of size m .

(a) Show that for any sample S ,

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_1 + \mathcal{H}_2) = \widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2).$$

(b) Show that the empirical Rademacher complexity of $\mathcal{H}$ for any sample S of size m can be bounded as follows:

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_1 \mathcal{H}_2) \leq \frac{3}{2} \left[\widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2) \right].$$

Hint: use the identity $x_1 x_2 = \frac{1}{4} [(x_1 + x_2)^2 - (x_1 - x_2)^2]$ valid for all $x_1, x_2 \in \mathbb{R}$ and Talagrand's contraction lemma.

2. Let $\mathcal{E}$ be the family of functions defined over $\mathcal{X}$ by $\mathcal{E} = \{x \mapsto \mathbb{1}_{x=z} : z \in \mathcal{X}\}$.

(a) Let S be a sample of size m with distinct points. Prove the inequality $\widehat{\mathfrak{R}}_S(\mathcal{E}) \leq \frac{1}{m}$.

(b) Let S be a sample of size m with a single point repeated m times. Prove the equality

$$\widehat{\mathfrak{R}}_S(\mathcal{E}) = \frac{1}{2m} \mathbb{E}_{\sigma} \left[\left| \sum_{i=1}^m \sigma_i \right| \right].$$

Show that $\widehat{\mathfrak{R}}_S(\mathcal{E}) \leq \frac{1}{2\sqrt{m}}$.

(c) 1[bonus question] Let n^* denote the largest number of repetitions of any point in the sample. Give an upper bound on $\widehat{\mathfrak{R}}_S(\mathcal{E})$ in terms of n^* .